



Adsorption of tetracycline hydrochloride onto ball-milled biochar: Governing factors and mechanisms

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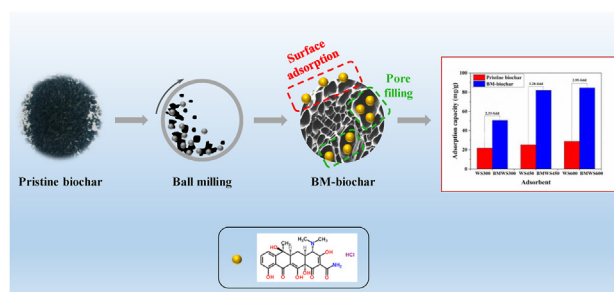
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HIGHLIGHTS

- Ball milling enhanced external specific surface area and mesoporous volume of biochar.
- Ball milling increased sorption ability of biochars to TCH.
- TCH adsorption by BM-biochars was controlled by surface adsorption and pore filling.
- TCH adsorption on BM-biochars was affected by pH and cation in solution.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 6 April 2020

Received in revised form

8 May 2020

Accepted 10 May 2020

Available online 11 May 2020

Handling Editor: Y Yeomin Yoon

Keywords:

Engineered biochar

Ball milling

Antibiotic

Tetracycline hydrochloride

Adsorption

ABSTRACT

Pristine and ball milled wheat stalk biochars pyrolysed at 300 °C, 450 °C, 600 °C were studied for tetracycline hydrochloride (TCH) adsorption from aqueous solution. Surface characteristics of ball milled biochar (BM-biochar) were significantly enhanced over their pristine counterparts. TCH adsorption occurred largely on external surface and by filling pores of biochars as evidenced by strong positive correlation between adsorption and external specific surface area (SSA), total pore volume, or mesoporous volume. A two-stage intra-particle diffusion model, limited by the TCH diffusion through the boundary liquid layer, well described TCH adsorption. Maximum TCH adsorption occurred at about pH = 6–8. While solution cations including Na⁺, K⁺ and Mg²⁺ subdued TCH adsorption as they competed for adsorption sites, Ca²⁺ promoted TCH adsorption due to formation of tetracycline-Ca²⁺ complexes. The best performing BM-biochar was the one pyrolysed at 600 °C with TCH adsorption amount of 84.54 mg/g. Therefore, this BM-biochar has the potential for TCH removal from aqueous solutions. And the research shed light on the management of organic contaminants in real wastewater by BM-biochar.

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1. Introduction

Antibiotics, the most important antibacterial agents against

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bacterial infections, are widely used in industries such as medicine, animal husbandry, and aquaculture to kill bacteria or inhibit their growth (Wang et al., 2020). However, antibiotics cannot be directly metabolized by humans or animals, and most of them are excreted in the form of original drugs or parent compounds to the environment (Singh et al., 2019). Long-term residual antibiotics may result in proliferation and transmission of drug-resistant bacterial flora and antibiotic-resistant genes (ARGs), which may, in turn, adversely affect the structure and function of ecosystems (Chu et al., 2020). In spite of the huge social and economic benefits, the environmental impact by residue antibiotics, especially the probable perennial adverse effects on human health and ecosystems is becoming a serious concern. Therefore, it is urgent to develop effective and economical remediation technology to control antibiotics in the environment.

Several treatment techniques including photocatalytic degradation (Gholami et al., 2020), adsorption (Manjunath et al., 2020), biodegradation (Zhi et al., 2019), and advanced oxidation (Anjali and Shanthakumar, 2019) have been studied for antibiotics removal from polluted water. Adsorption is perhaps the most commonly used technology in wastewater treatment to remove both inorganic and organic pollutants (Zhang et al., 2017a; Wang et al., 2018b). The technical effectiveness and economic efficiency of adsorption technology depend on the adsorbents selected for usage. The existing literature has documented characteristics of numerous adsorbents for antibiotics such as activated carbon (Ahmed, 2017), graphene-based materials (Li et al., 2019), biochar (Huang et al., 2019; Xiang et al., 2020a), microalgae (Leng et al., 2020), polyurethane polymer (Okoli and Ofomaja, 2019), carbonaceous nanochlorapatites (Deng et al., 2019), and clay minerals (Awad et al., 2019). Among those, biochar is considered a highly effective adsorbent based on its high SSA, porous structure and abundant surface functional groups (SFG) (Wang et al., 2018a). Biochar technology has been studied for applications in wastewater and stormwater treatment, soil remediation, and waste gas reduction (Zhang et al., 2017b; Khan et al., 2019; Wang and Wang, 2019).

To further enhance the adsorption performance of biochar for broadened environmental applications, several physical or chemical modification techniques were studied based on an assessment of the improvement in physicochemical characteristics of the modified biochar over the pristine biochar (Wang and Wang, 2019; Zhang et al., 2019b). Ball milling is a physical modification method using high levels of mechanical energy to grind the pristine biochar into nanoscale particles, thereby activating chemical enhancement without involving any chemical reagents (Naghdi et al., 2017). This technique has attracted much attention with numerous studies examining the physical and chemical features of the new materials (Lyu et al., 2018b; Ai et al., 2019; Zhang et al., 2019a), as well as the adsorption performance on a wide spectrum of pollutants such as methylene blue (Lyu et al., 2018b), heavy metals (Lyu et al., 2018a), ammonium (Qin et al., 2019), and carbon dioxide (Xu et al., 2019). However, only few studies investigated the mechanism and effectiveness of BM-biochar for removal of tetracycline hydrochloride (TCH) from aqueous solution.

TCH is a hydrochloride salt of tetracycline, a broad-spectrum naphthalene antibiotic synthesized with chlortetracycline, which has been detected in rivers, lakes, sediments, soils, and even groundwater (Luo et al., 2011; Bhattacharyya et al., 2019). This study examines how pristine biochar and BM-biochar adsorb TCH. Specifically, the objectives of this work are to (1) compare the physicochemical properties of pristine biochar and BM-biochar, (2) evaluate the adsorption characteristics of TCH on pristine biochar and BM-biochar, (3) analyze the influences of solution pH and cations on TCH adsorption, and (4) elucidate the governing mechanisms of TCH adsorption.

2. Material and methods

2.1. Materials

Wheat stalk (WS) was collected from the farmland of Xuzhou, China. Chemical reagents (analytical grade) including TCH ($C_{22}H_{25}ClN_2O_8$, purity > 99.0%), NaOH, HCl, NaCl, KCl, $MgCl_2$, and $CaCl_2$. TCH was purchased from Taize Jiaye Co., Ltd., and other chemicals were bought from Sinopharm Chemical Reagent Co. Ltd.. Solutions were prepared using deionized water (18.2 M Ω).

2.2. Biochar production

WS sample was washed, oven dried and cut into 1 cm in length, and then pyrolysed at 300 °C, 450 °C, and 600 °C separately for 4 h in a tube furnace (N_2 flow rate was 50 mL/min, and the heating rate was 15 °C/min). Obtained biochars were rinsed with deionized water, dried at 105 °C overnight, sieved for fraction of 0.15–0.45 mm, and stored for further tests and ball milling. These pristine biochars were labeled as WS300, WS450, and WS600.

BM-biochars were produced in a planetary ball mill machine (biochar/ball mass ratio was 1:100). Machine rotation was set at 300 rpm in ambient air for 12 h with direction altered every 3 h. Obtained BM-biochar, labeled as BMWS300, BMWS450, BMWS600, were stored in desiccators for further experiments.

2.3. Characterization

SSA and pore characteristics of pristine biochar and BM-biochar were tested and calculated according to nitrogen adsorption-desorption isotherms. Morphology and structure of biochars were analyzed via SEM (HITACHI SU-8010 Scanning Microscope, Japan). SFG were detected by FTIR (Nicolet iS10, Thermo Nicolet Corporation USA). Total acidity was assessed using a Boehm titration method (Tsechansky and Graber, 2014).

2.4. Adsorption experiments

TCH adsorptive experiments were executed to evaluate the adsorption performance of pristine biochar and BM-biochar. The adsorption kinetic and isotherm experiments were conducted along with determination of the impacts of pH, initial cation type and concentration on TCH adsorption.

For adsorption kinetic experiments, 200 mg of each biochar sample was added to 1 L TCH solution (25 mg/L) in a brown bottle. All bottles were added the magnetic rotor, placed on a constant temperature magnetic heating stirrer, and sampled at different times (0.5 h, 1 h, 2 h, 3 h, 5 h, 10 h, 12 h, 24 h, 36 h, 48 h, 72 h, 96 h, 120 h, 144 h and 168 h) at 25 °C.

For adsorption isotherm experiments, each biochar sample (10 mg) was added to 50 mL TCH solutions (TCH aqueous concentration were 5 mg/L, 10 mg/L, 15 mg/L, 20 mg/L, 25 mg/L and 30 mg/L, respectively) in a plastic centrifuge tube. The mixture was shaken for 168 h at 25 °C until reached the adsorbed balance.

To determine the effect of initial solution pH on adsorption, each biochar sample (10 mg) was added to TCH solutions (initial TCH aqueous concentration was 25 mg/L) with pH of 4.0, 6.0, 8.0, 10.0, and 11.0, respectively. For the influence of cation type and concentration on the adsorption experiments, NaCl, KCl, $MgCl_2$ and $CaCl_2$ were introduced to the TCH solutions to reach a concentration range of 0.01–0.1 mol/L, the initial pH value of solution was 7.0. The experimental procedures of the isotherm experiment were followed.

All solution samples were filtered with a 0.22 μm syringe filter, and TCH concentration in filtrate was detected by UV

spectrophotometer at 356 nm. The desired solution pH values were adjusted by using HCl (0.05 mol/L) and NaOH (0.05 mol/L) solutions. The pH values monitored before and after the adsorption experiment, exhibited little difference (<5%).

All adsorption experiments were executed in triplicates and average values were calculated. Blank tests without biochar were performed to determine the influence of soluble substances released from biochar.

3. Results and discussion

3.1. Characterization of biochars

BET-SSA, external SSA (the sum of all external surfaces per gram biochar), total pore volume (TPV) and mesoporous volume of pristine biochars increased with increasing pyrolysis temperature, while the Barrett-Joyner-Halenda (BJH) pore diameter decreased, especially for biochars derived at 600 °C (Table 1). This phenomenon is mainly associated with increased decomposition of biomass and release of volatile materials at high pyrolysis temperatures (Xiang et al., 2020b). After ball milling, BET-SSA, external SSA, TPV and mesoporous volume significantly increased compared to those of corresponding pristine biochars. BMWS450 showed the most dramatic increase in external SSA (by 14 folds) and mesoporous volume (by 8.5 folds). This supports that ball milling generates ultrafine biochar particles with abundant open pores, thereby increasing its surface area and mesoporous volume (Lyu et al., 2017a; Xiang et al., 2020b).

Surface morphologies of pristine biochars and BM-biochars were examined by SEM (Fig. 1). Surface morphologies of pristine biochars (Fig. 1a and b, c) commonly exhibited the layered structure and preliminary pores with porous, rough micro-scaled particles on the surface (Tan et al., 2019). After ball milling, the layered structure of pristine biochar was turned into irregular ultrafine particles (Fig. 1d and e, f), and a large number of nanoparticles less than 100 nm were obtained.

FTIR spectra of pristine biochars and BM-biochars showed adsorptive peaks at 792 cm⁻¹, 1100 cm⁻¹, 1370 cm⁻¹, 1564 cm⁻¹, 2919 cm⁻¹, and 3425 cm⁻¹ (Fig. 2), corresponding to the aromatic C-H (Lian et al., 2013), C-O (Lian et al., 2013), phenolic O-H (Lyu et al., 2017b), aromatic C=C (Chen et al., 2015), aliphatic C-H (Zhang and Shen, 2019), and -OH (Shen and Zhang, 2019). For BM-biochars, the bending vibration of aromatic C-H, C-O, phenolic O-H, and aromatic C=C were significantly enhanced. It indicated that more functional groups were formed on biochar surface during the ball milling process. Because ball milling process might introduce oxygen-containing functional groups to the surface of the biochars (Lyu et al., 2017a). These SFG can improve the adsorption performance of the biochars via electrostatic interaction, surface complexation between the SFG and adsorbates (Lyu et al., 2018a, 2018b).

Total acidity of pristine biochars and BM-biochars decreased as pyrolysis temperatures increased (Table 1). This is because at higher temperatures, some SFG such as carboxylic, phenolic

hydroxyl and carbonyl are decomposed (Ding et al., 2016). Ball milling increased the total acidity (Table 1), indicative of increased oxygen-containing functional groups, which are the main source of surface acidity of carbon materials (Li et al., 2011; Lyu et al., 2018b). This is consistent with the FTIR spectra in that enhanced adsorptive peaks (C-O, phenolic O-H and -OH) of oxygen-containing function groups were observed in BM-biochars.

3.2. Adsorption capacities

Under the same experimental conditions, TCH adsorbed on pristine biochars ranged between 21.67 and 28.64 mg/g (Fig. 3), lower than that of BM-biochars (50.58–84.55 mg/g). And the biochar derived from higher pyrolysis temperature showed the better adsorption performance for TCH. It might be due to higher pyrolysis temperature conducted to the development of biochars' pore structure.

Ball milling lead to about 3-fold increase in TCH adsorption, indicative of a viable engineering means to improve adsorptive properties of biochars. Because adsorption capacity is commonly determined by the surface area and/or pore volume of adsorbents (Inyang et al., 2015; Tan et al., 2019), we conducted linear correlation of TCH adsorption capacity against the following parameters: (1) BET-SSA, (2) t-Plot external SSA, (3) TPV, and (4) Mesoporous volume (Fig. 4). While only marginal correlation was found between BET-SSA and TCH adsorption capacity ($R^2 = 0.31$), strong positive correlation existed between external SSA, TPV, or Mesoporous volume and TCH adsorption capacity ($R^2 = 0.85$, $R^2 = 0.80$, and $R^2 = 0.81$, respectively). This correlation analysis illustrates that TCH adsorption occurred largely on the external surface and pores of biochars (Zeng et al., 2019).

3.3. Adsorption kinetics and isotherm

TCH adsorption kinetics were explored with the pseudo-1st order, pseudo-2nd order and intra-particle diffusion (IPD) models. Fig. 5 and Table 2 show the pseudo-1st order and pseudo-2nd order fitting curves and parameters. Both models fitted the experimental data well ($0.90 \leq R^2 \leq 0.99$). Pseudo-2nd order model described TCH adsorption better than the pseudo-1st model, suggesting that TCH adsorption on biochars was likely heterogeneous and followed the binuclear surface adsorption mechanism (Fang et al., 2017). In addition, the theoretical q_e values of BM-biochars were higher than for pristine biochars, e.g., the theoretical q_e of BMWS600 was 2.94-fold higher than for WS600. Based on analysis of biochar properties, external SSA and TPV of BMWS600 were 6.92-fold and 1.95-fold larger than that of WS600, indicating that surface adsorption and pore filling were attributable to the higher q_e of BM-biochar than corresponding pristine biochar. Meanwhile, the enhanced SFG of BM-biochar may also promote chemical interactions with TCH partially explaining the higher q_e value of BM-biochars than for pristine ones.

Fig. 6 and Table 3 show the fit between experimental data and

Table 1
Characteristics of pristine and ball-milled wheat stalk biochars.

	Sample	BET-SSA (m ² /g)	t-Plot external SSA (m ² /g)	TPV (cm ³ /g)	Mesoporous volume (cm ³ /g)	BJH pore diameter(nm)	Total acidity (mmol/g)
Pristine biochar	WS300	3.90	3.46	0.0174	0.0172	4.84	1.48
	WS450	33.69	9.97	0.0544	0.0398	3.00	0.51
	WS600	211.56	14.72	0.1370	0.0473	2.28	0.01
Ball-milled biochar	BMWS300	13.50	13.50	0.0706	0.0706	4.03	1.62
	BMWS450	139.89	139.89	0.3366	0.3366	2.27	1.34
	BMWS600	257.50	101.90	0.2678	0.1966	2.41	0.48

Note: SSA = Specific surface area, TPV = Total pore volume, and BJH = Barrett-Joiner- Halenda.

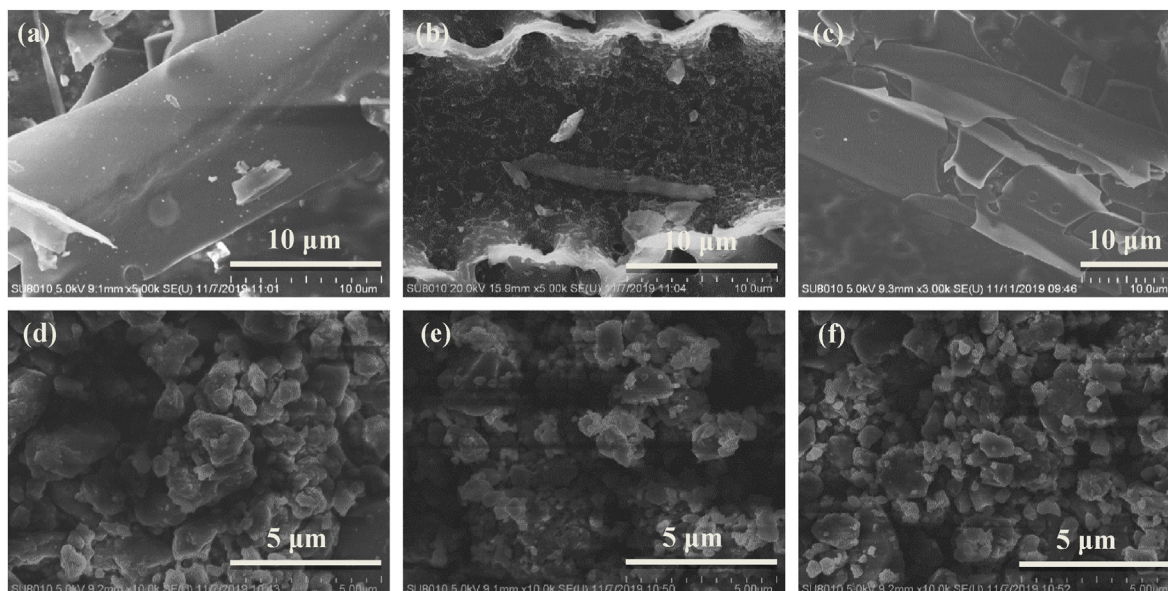


Fig. 1. Scanning electron micrographs (SEM) images of (a) WS300, (b) WS450, (c) WS600, (d) BMWS300, (e) BMWS450, and (f) BMWS600.

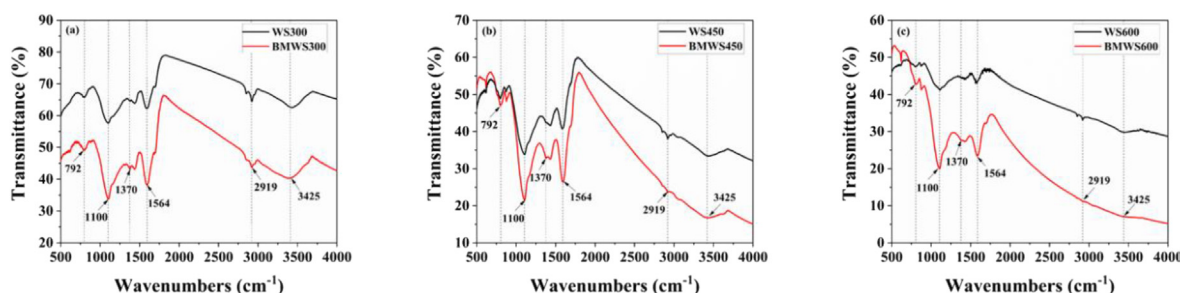


Fig. 2. FTIR spectra of pristine and ball-milled wheat stalk biochars pyrolysed at (a) 300 °C, (b) 450 °C, (c) 600 °C.

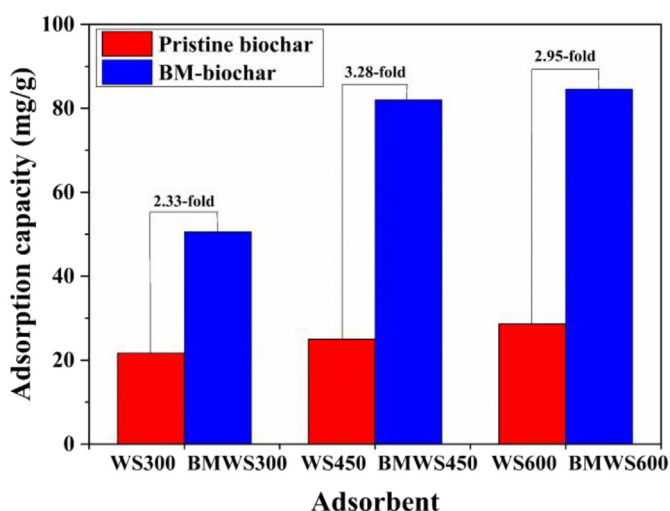


Fig. 3. Adsorption capacity of TCH on pristine and ball-milled wheat stalk biochars.

IPD model parameters. The R^2 values of IPD model ranged from 0.71 to 0.99, indicating that TCH adsorption was largely governed by the diffusion process (Zeng et al., 2019; Qu et al., 2020). Two distinct diffusion stages were identified, whereby the linear relationship

between TCH adsorptive amount and $t^{0.5}$ differed mainly in IPD rate (k_{ip}). The first stage had a much larger IPD rate constant (k_{1p}), indicative of the external diffusion dominated process. The diffusion time during this stage was, in general, less than 9 h. TCH adsorption was likely governed by molecule diffusion and film diffusion (Zeng et al., 2019). The second stage had a gentler slope, suggesting that the internal diffusion process dominated TCH transfer from its external surface into the pores of the biochar matrices. Also note that the diffusion rate (k_{1p}) of BM-biochars (13.93–19.87) was roughly 3-fold higher than that of pristine ones (4.57–7.40) in Phase 1. This is likely attributed to the enlarged external SSA and TPV of BM-biochars, which provided more adsorption sites and pore filling volume for TCH. In Phase 2, the diffusion rate (k_{2p}) of BM-biochars was not much higher than that of pristine ones, indicating that the intra-particle diffusion rate decreased after sufficient surface adsorption and pore filling of TCH.

The intercept C in IPD (Table 3) generally indicates the range of the boundary layer thickness, that is, the larger the intercept, the greater the boundary layer effect (Mckay et al., 1980). Larger values of C_2 than C_1 suggest that the diffusion boundary layer during stage 2 became thicker. These results also indicate that the film diffusion of TCH molecules transferred through the boundary liquid layer was the most important limiting step of the adsorption process.

Adsorption isotherm results of TCH by Langmuir and Freundlich models were shown in Fig. 7 and Table 4. Langmuir model described the experimental data better than Freundlich model.

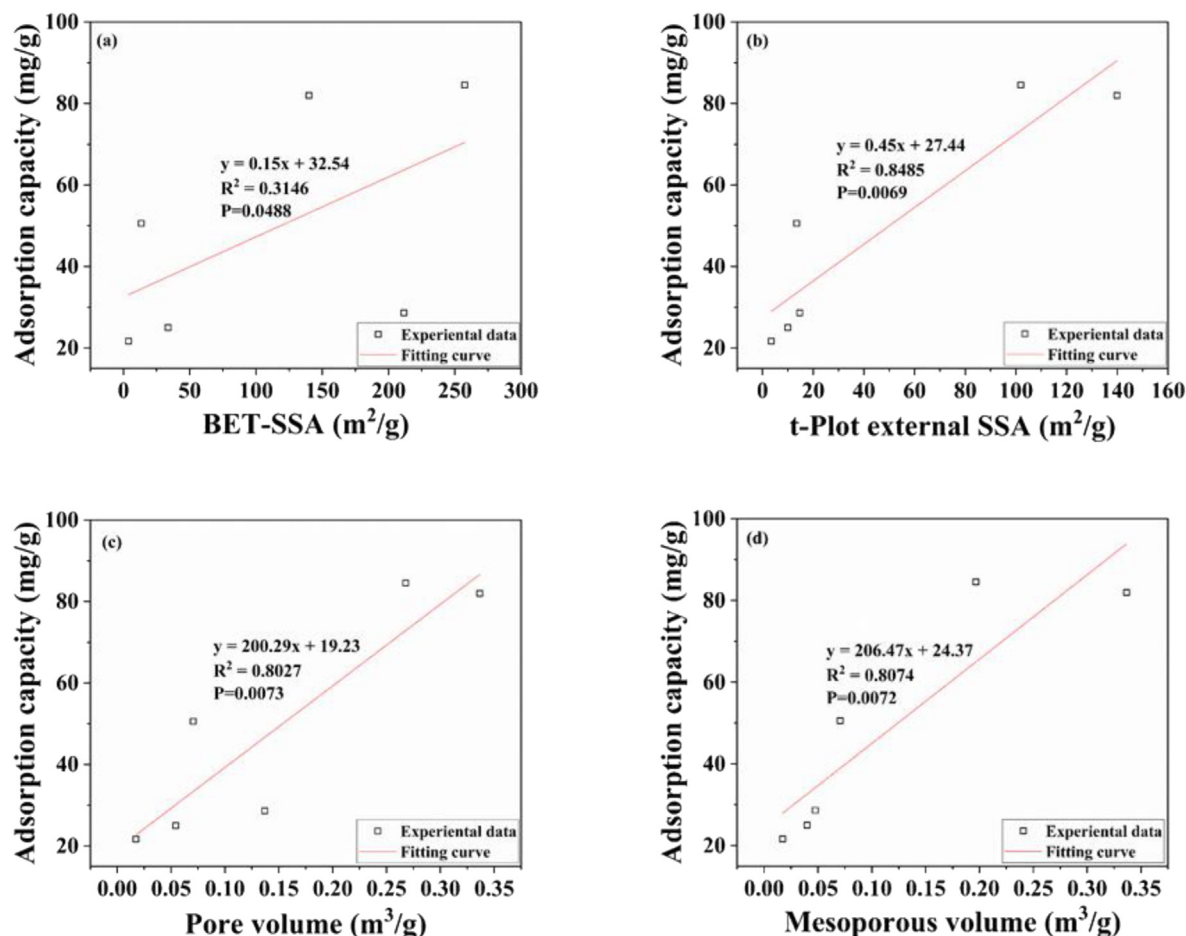


Fig. 4. Linear correlation analysis of (a) BET-SSA vs. TCH adsorption capacity, (b) t-Plot external SSA vs. TCH adsorption capacity, (c) TPV vs. TCH adsorption capacity, (d) Mesoporous volume vs. TCH adsorption capacity.

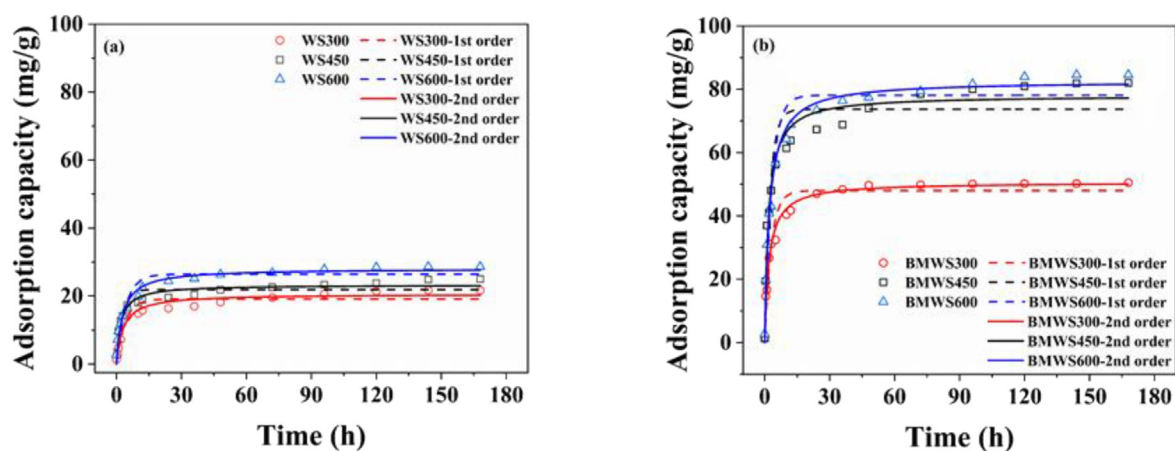


Fig. 5. Adsorption kinetics of TCH on pristine (a) and ball-milled (b) biochars derived from wheat stalk.

These results revealed that adsorption of TCH onto pristine biochars and BM-biochars was likely governed by the monolayer molecular behavior. The q_{max} (Langmuir model) and K_F (Freundlich model) of BM-biochars were higher than that of pristine biochars, indicative of improved TCH adsorption capacity by ball milling. This can again be attributed to the enlarged external SSA and TPV.

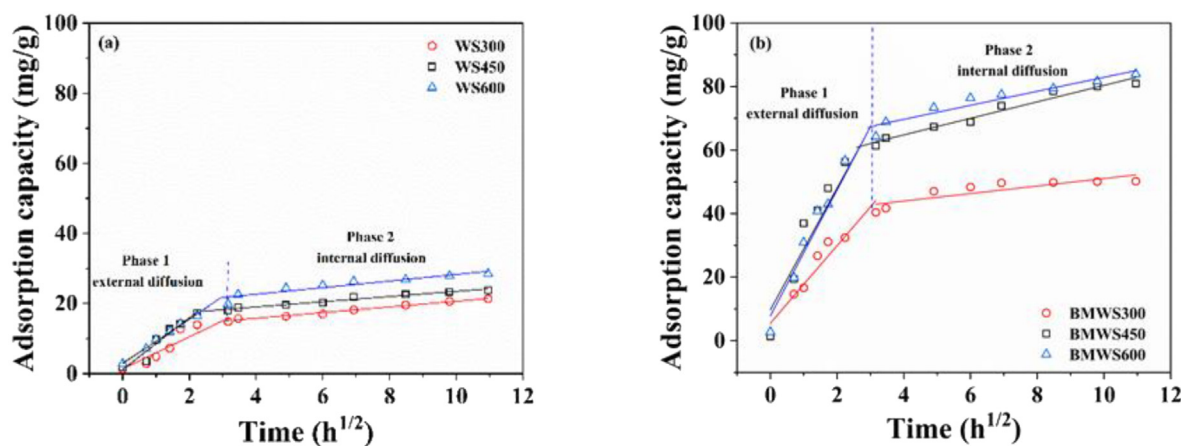
3.4. Effect of pH

pH effects on adsorption of TCH onto biochars changed after ball milling (Fig. 8). For the pristine biochars, increase in TCH adsorption was obvious with the initial pH value increased from 4 to 6. TCH adsorption did not change much with further increase in pH to 8, and for WS600 TCH adsorption even increased. Further increase in

Table 2

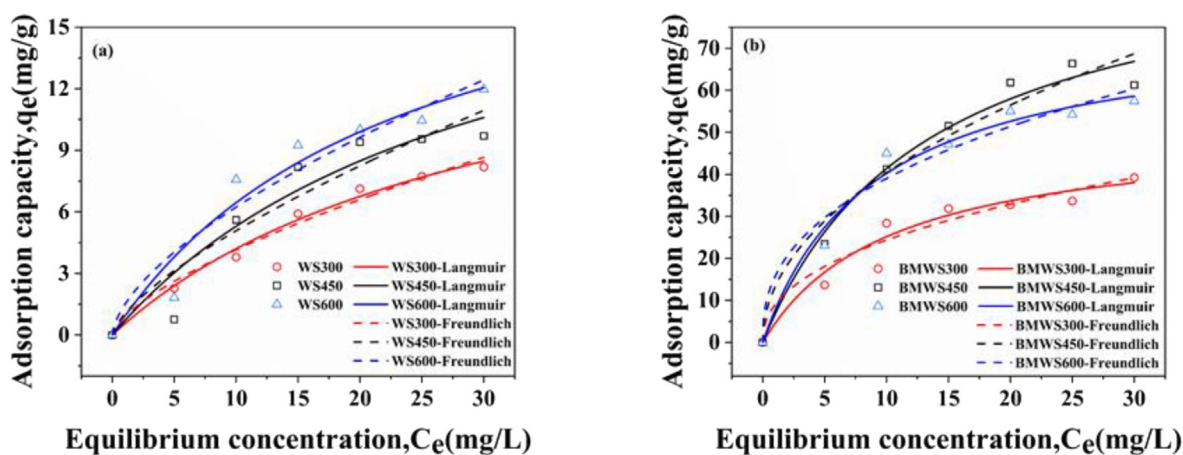
Adsorption kinetics parameters of TCH on pristine and ball-milled wheat stalk biochars.

Model kinetics	Pseudo-1st order: $q_t = q_e(1 - e^{-k_1 t})$			Pseudo-2nd order: $q_t = \frac{t \cdot k_2 \cdot q_e^2}{1 + q_e \cdot t \cdot k_2}$			Experiment sorption capacity $Q/(mg/g)$
	$q_e/(mg/g)$	$k_1/(min^{-1})$	R^2	$q_e/(mg/g)$	$k_2/(g/(mg \cdot min))$	R^2	
WS300	19.10	0.2581	0.9135	20.59	0.0172	0.9580	21.67
WS450	21.86	0.3803	0.9063	23.25	0.0222	0.9595	25.00
WS600	26.44	0.2387	0.9091	28.08	0.0127	0.9681	28.64
BMWS300	47.97	0.3342	0.9376	50.03	0.0087	0.9857	50.58
BMWS450	73.69	0.4017	0.9016	78.01	0.0074	0.9665	81.97
BMWS600	78.08	0.3067	0.9385	82.63	0.0055	0.9859	84.55

**Fig. 6.** Kinetic plots of intra-particle diffusion model of TCH on pristine (a) and ball-milled (b) biochars derived from wheat stalk.**Table 3**

Kinetic plots of intra-particle diffusion model parameters of TCH on pristine and ball-milled wheat stalk biochars.

Biochar	Phase	$k_{ip}(mg/(g \cdot min^{0.5}))$	$C_i(mg/g)$	R_i^2	Biochar	Phase	$k_{ip}(mg/(g \cdot min^{0.5}))$	$C_i(mg/g)$	R_i^2
WS300	1	4.5747	1.3614	0.8759	BMWS300	1	13.9328	6.2400	0.9246
	2	0.8013	12.6143	0.9826		2	1.3524	44.7111	0.7065
WS450	1	7.4018	1.1532	0.9247	BMWS450	1	19.1908	9.6650	0.8709
	2	0.7435	16.0338	0.9707		2	2.5962	54.3939	0.9607
WS600	1	6.3004	2.9345	0.9924	BMWS600	1	19.8676	7.7864	0.9441
	2	0.9425	18.8641	0.8614		2	2.2067	60.8338	0.9066

Note: IPD equation is $q = k_{ip} \times t^{0.5} + C_i$, q is the adsorptive amount, k_{ip} is the IPD rate constant, t is time, i is the stages of the IPD model, i is 1 or 2.**Fig. 7.** Adsorption isotherms of TCH on pristine (a) and ball-milled (b) wheat stalk biochars at 25 °C.

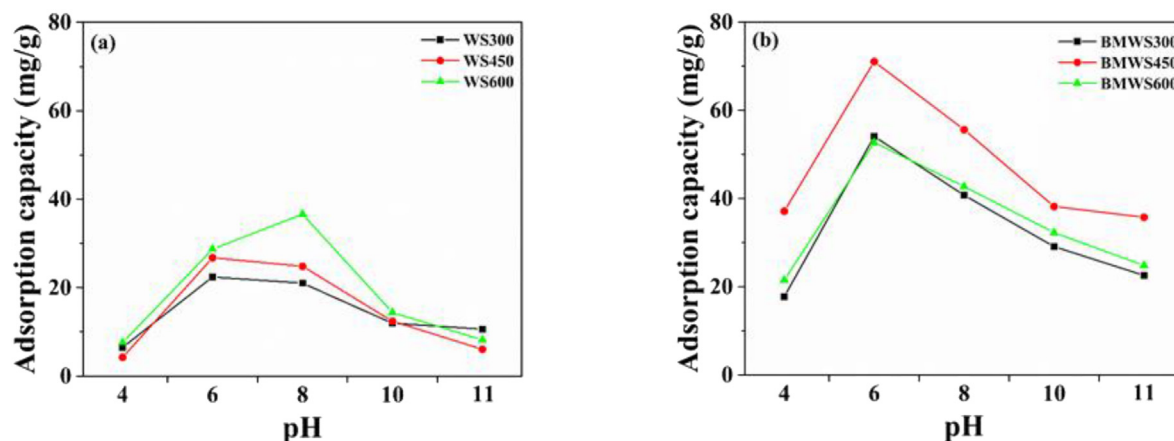
pH from 8 to 11 reduced TCH adsorption. For BM-biochars, increase in TCH adsorption occurred only when the initial pH increased from

4 to 6. Further increase in pH reduced TCH adsorption. Also note that WS600 and BMWS450 had the best adsorption performance

Table 4

Adsorption isotherm parameters of TCH on pristine and ball-milled wheat stalk biochars.

Sample	Langmuir: $q_e = bq_{max}C_e/(1+bC_e)$			Freundlich: $q_e = K_f C_e^{1/n}$		
	$q_{max}/(\text{mg/g})$	$b/(\text{L/mg})$	R^2	$K_f/(\text{mg}^{(1-1/n)} \cdot \text{L}^{1/n}/\text{g})$	n	R^2
WS300	17.19	0.0324	0.9763	0.8899	1.4953	0.9678
WS450	21.19	0.0333	0.9034	1.0188	1.4329	0.8773
WS600	21.29	0.0434	0.9412	1.4695	1.5932	0.9176
BMWS300	51.04	0.0971	0.9677	8.9928	2.3101	0.9491
BMWS450	96.69	0.0747	0.9776	13.2649	2.0681	0.9559
BMWS600	75.95	0.1127	0.9759	15.6765	2.5238	0.9535

**Fig. 8.** Effect of pH value on the TCH adsorption on pristine (a) and ball-milled (b) wheat stalk biochars.

under a wide range of solution pH among the pristine biochars and BM-biochars, respectively.

The pH effect on TCH adsorption was mainly related to the fact that TCH is an amphoteric molecule with multiple ionizing functional groups. In solution, TCH exists as TCH_3^+ at pH less than 3.3, TCH_2^0 with pH between 3.3 and 7.7, and $\text{TCH}^-/\text{TC}^{2-}$ with pH greater than 7.7 (Wan et al., 2019; Zeng et al., 2019). Because at lower pH the surface of biochars and associated functional groups were commonly positively charged (Lyu et al., 2018b; Wan et al., 2019), electrostatic repulsion would inhibit the TCH adsorption on the biochars. With increase of pH, TCH was gradually deprotonated with increase in TCH_2^0 , $\text{TCH}^-/\text{TC}^{2-}$ species. The electrostatic interaction between TCH and biochars were also enhanced, leading to an increase in adsorptive amount of TCH. Meanwhile, surfaces of biochars were also likely negatively charged at higher pH (Wan et al., 2019). The electrostatic repulsion between TCH and negatively charged biochars reduced the adsorption capacity of TCH. The different pH effect after ball milling can also be understood in a context of changed SFG and electrostatic properties of these biochars.

3.5. Effect of coexisting cation type and strength

The presence of different ions (Na^+ , K^+ , Mg^{2+} , Ca^{2+}) in solution and their concentrations affected TCH adsorption on biochars (Fig. 9). With the increase of Na^+ , K^+ and Mg^{2+} initial concentrations in aqueous solution, TCH adsorption on both pristine and ball-milled biochars decreased. This is because cation added in TCH solution, such as Na^+ , K^+ and Mg^{2+} , can occupy active adsorption sites on the biochar's surface and generate competitive adsorption against TCH, thereby reducing the electrostatic attraction between biochar and TCH (Zeng et al., 2019). In contrast, with the increase of Ca^{2+} initial concentration, TCH adsorption on both pristine biochar

and BM-biochar gradually increased. It is well known that the oxygen-containing functional groups such as hydroxyl in TCH are ionizable with strong complexation ability with Ca^{2+} (mainly $-\text{OH}$ and $-\text{C}=\text{O}$). Therefore, the formation of tetracycline- Ca^{2+} complexes promotes the adsorption of TCH by pristine and ball-milled biochars.

4. Conclusions

Ball milling technology significantly improved the surface characteristics of pristine wheat stalk biochars pyrolysed at varying temperatures. The adsorption capacity of TCH onto BM-biochars increased by about three folds over that of their pristine counterparts. The enhanced TCH adsorption capacities were positively correlated with increased external SSA, total pore volume and mesoporous volume achieved by ball milling. The kinetics of TCH adsorption onto biochars was governed by the intra-particle diffusion process, and the isotherms followed the Langmuir model. Because TCH's amphoteric molecule contains ionizing functional groups, solution pH and cations affected its adsorption onto biochars. The optimal pH was at about 6–8. Cations such as Na^+ , K^+ and Mg^{2+} negatively affected TCH adsorption because they competed adsorption sites with TCH. However, Ca^{2+} enhanced TCH adsorption due to formation of tetracycline- Ca^{2+} complexes. The best performing BM-biochar was the one pyrolysed at 600 °C, which had a maximum TCH adsorption amount of 84.54 mg/g. This BM-biochar is a promising remediation absorbent for practical applications to treat TCH contaminants in the environment. Therefore, considering more widespread application of BM-biochar, more research such as removal of coexist organic pollutants in wastewater and the stability of BM-biochar should be studied in the future.

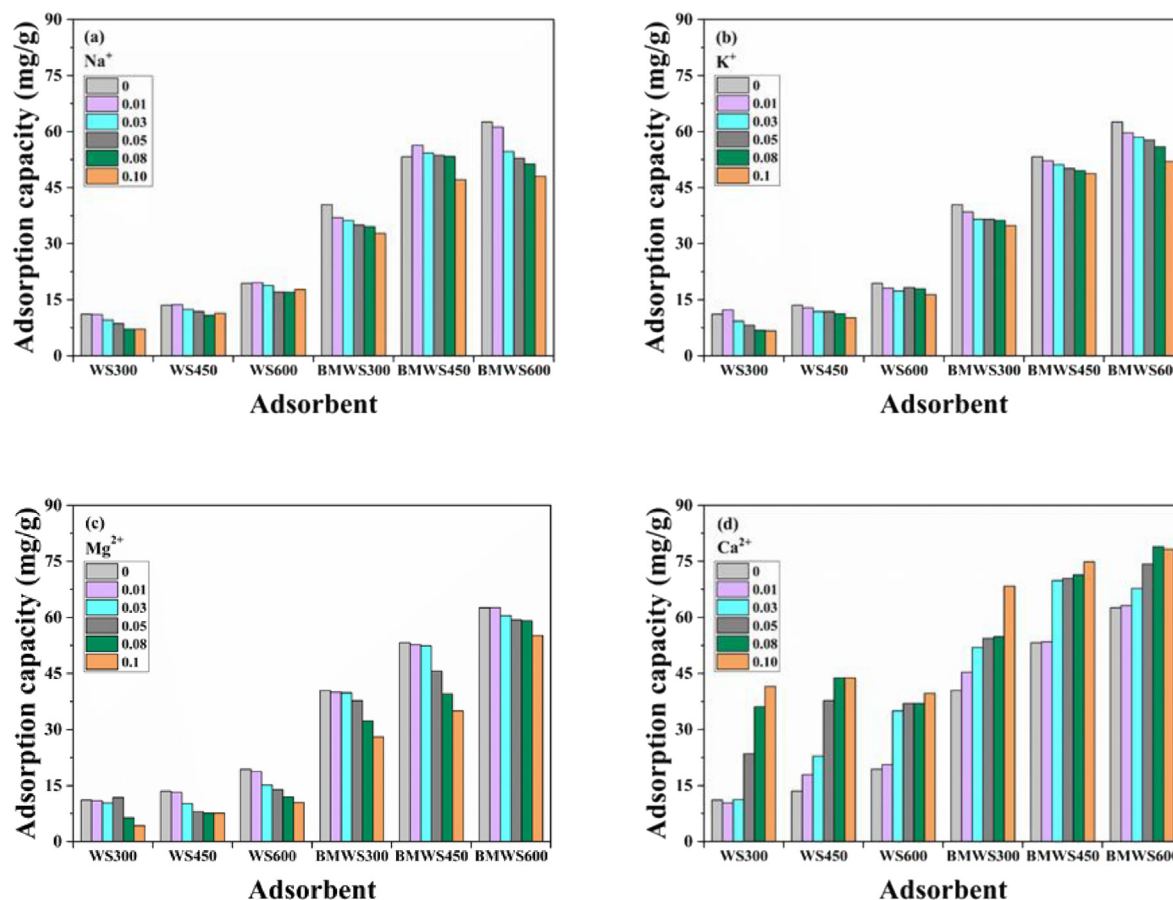


Fig. 9. Effect of cationic concentrations (0–0.1 mol/L) on the TCH adsorption onto pristine and ball-milled biochars at 25 °C, (a) Na⁺ (b) K⁺ (c) Mg²⁺ (d) Ca²⁺.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Wei Xiang: Conceptualization, Methodology, Formal analysis, Investigation, Writing - original draft. **Yongshan Wan:** Formal analysis, Writing - review & editing. **Xueyang Zhang:** Conceptualization, Methodology, Formal analysis, Writing - review & editing. **Zhenzhen Tan:** Methodology, Formal analysis, Writing - review & editing. **Tongtong Xia:** Methodology, Formal analysis, Writing - review & editing. **Yulin Zheng:** Methodology, Formal analysis, Writing - review & editing. **Bin Gao:** Conceptualization, Methodology, Formal analysis, Writing - review & editing.

Acknowledgments

This work was partially supported by the Natural Science Foundation of the Jiangsu Higher Education Institutions (Grant No. 18KJA610003), the Science and Technology Plan Projects of Xuzhou City (Grant No. KC18150, KC16SS091), and Xuzhou University of Technology (Grant NO. XKY2018136). The views expressed in this article are those of the authors and do not necessarily reflect the views or policies of the U.S. Environmental Protection Agency.

Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.chemosphere.2020.127057>.

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